

Name: Samson Kitigo

Student ID: 618666

Selected Significant Use-Cases

- UC-01 – View Live Match Updates: establish an SSE subscription, receive a connected event and current fixture detail, receive updates produced by the independent fixture poller, and restore current state from a fresh snapshot after reconnect.
- UC-02 – Follow a Team and Configure Alerts: search for a valid team, save the authenticated user-to-team association, independently save the user's global notification toggles, and register an Expo push device when necessary.
- UC-03 – Discover Official Tickets: resolve a fixture into Ticketmaster search parameters, show Ticketmaster event results, and route the fan through the backend buy endpoint to an Impact affiliate or direct Ticketmaster URL without processing payment or fulfillment.

1. View Live Match Updates

Primary actor: Fan **Trigger:** Fan opens a fixture **Success outcome:** Fresh snapshot plus live SSE updates

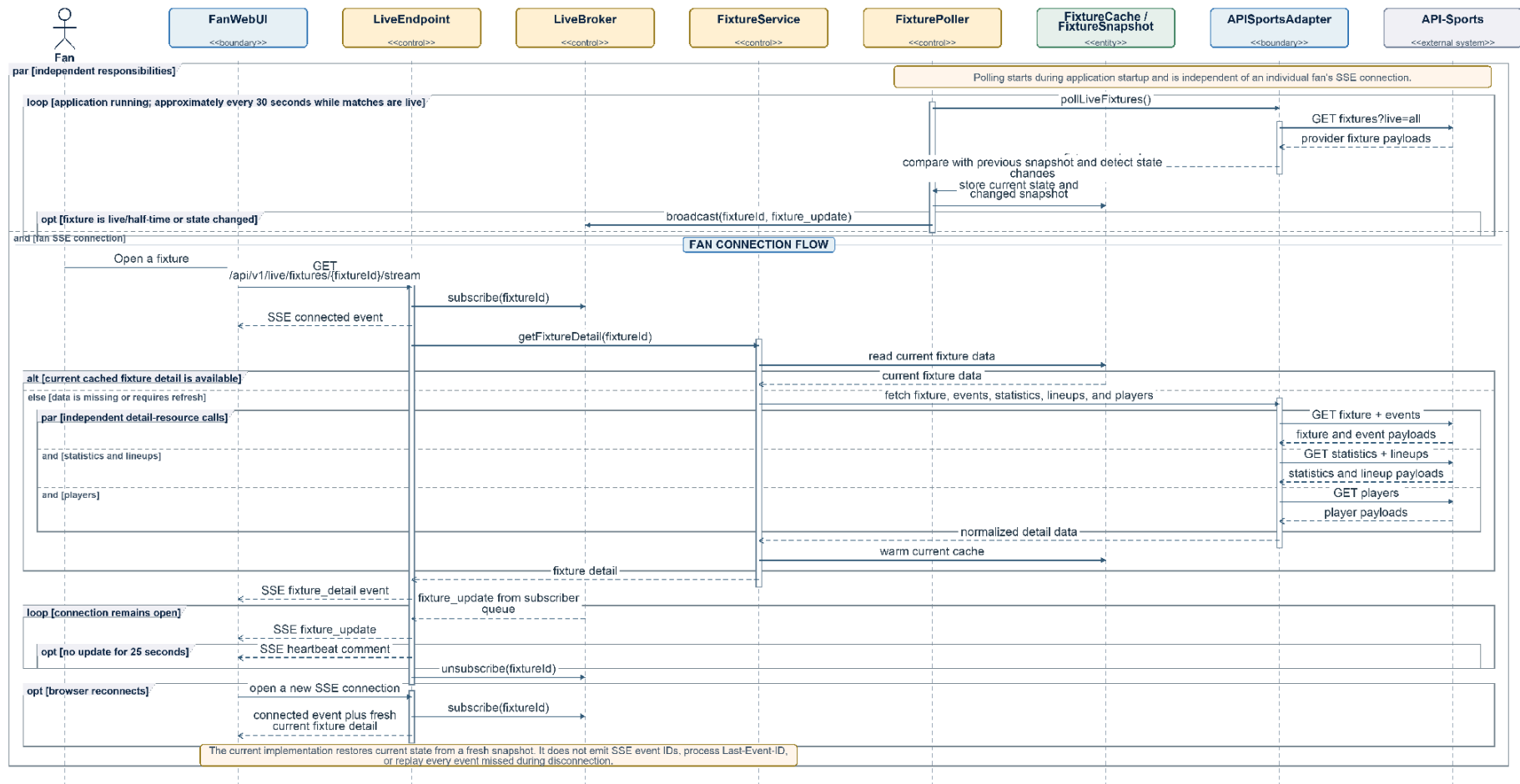


Figure 1. UML sequence diagram for view live match updates.

Analysis classes: Boundary – FanWebUI and APISportsAdapter; Control – LiveEndpoint, LiveBroker, FixtureService, and FixturePoller; Entity – FixtureCache and FixtureSnapshot.

2A. Search for and Follow a Team

Primary actor: Fan **Trigger:** Fan searches and selects teams **Precondition:** Fan is authenticated for follow request

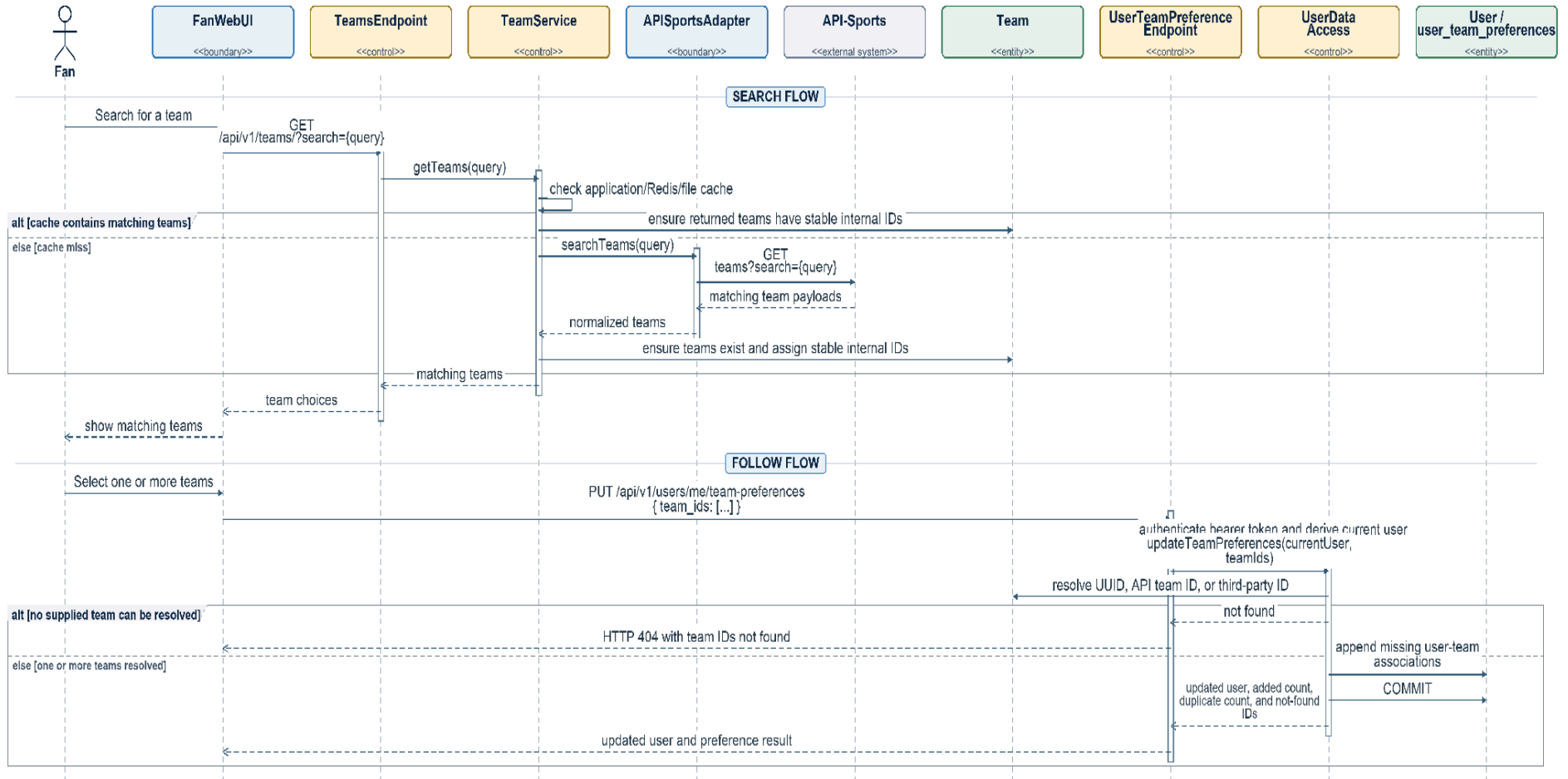


Figure 2A. UML sequence diagram for searching for and following a team.

Analysis classes: Boundary – FanWebUI and APISportsAdapter; Control – TeamsEndpoint, TeamService, UserTeamPreferenceEndpoint, and UserDataAccess; Entity – Team, User, and user_team_preferences.

2B. Configure Notification Preferences and Push Delivery

Primary actor: Fan **Trigger:** Fan changes global notification settings **Success outcome:** Preferences and optional Expo device are saved separately

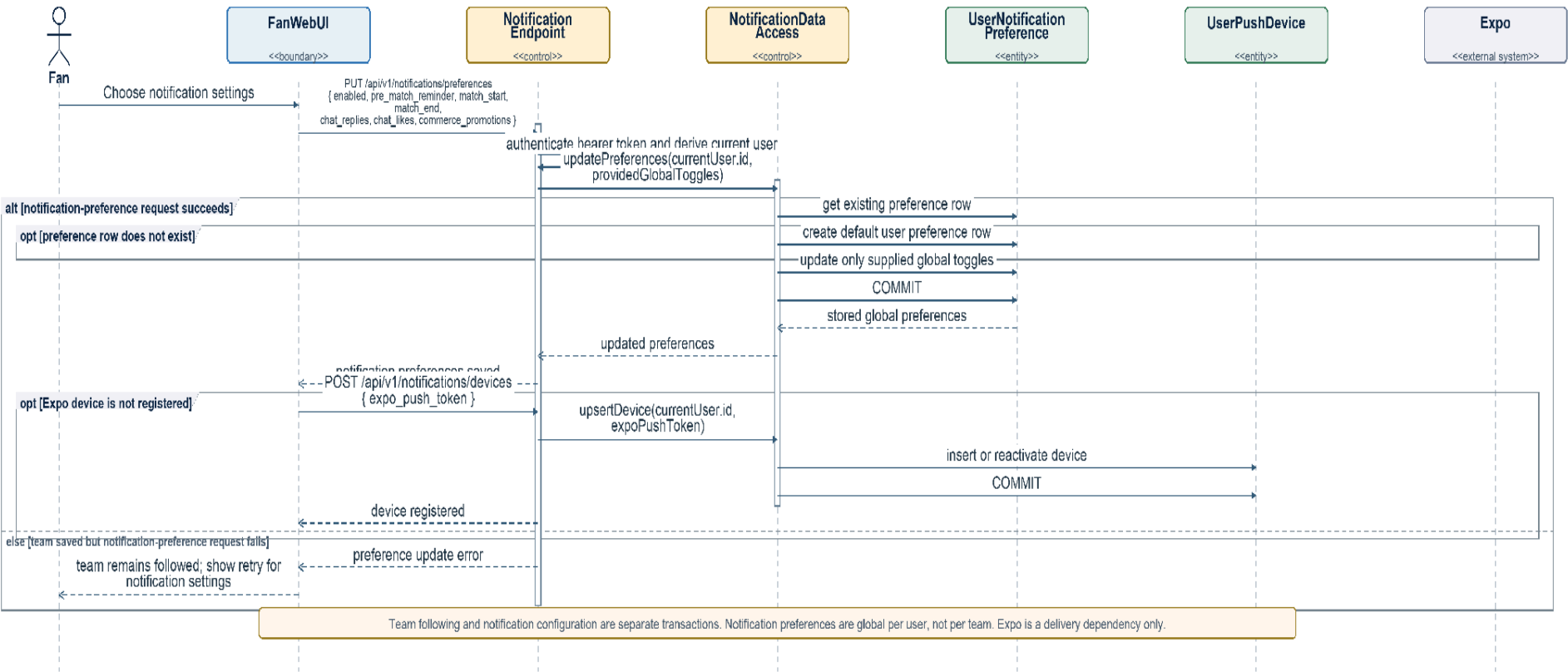


Figure 2B. UML sequence diagram for configuring notification preferences and push delivery.

Analysis classes: Boundary – FanWebUI and APISportsAdapter; Control – TeamsEndpoint, TeamService, UserTeamPreferenceEndpoint, UserDataAccess, NotificationEndpoint, and NotificationDataAccess; Entity – Team, User, user_team_preferences, UserNotificationPreference, and UserPushDevice.

3. Discover Official Tickets

Primary actor: Fan **Trigger:** Fan requests tickets for a fixture **Success outcome:** Ticketmaster results and safe external purchase redirect

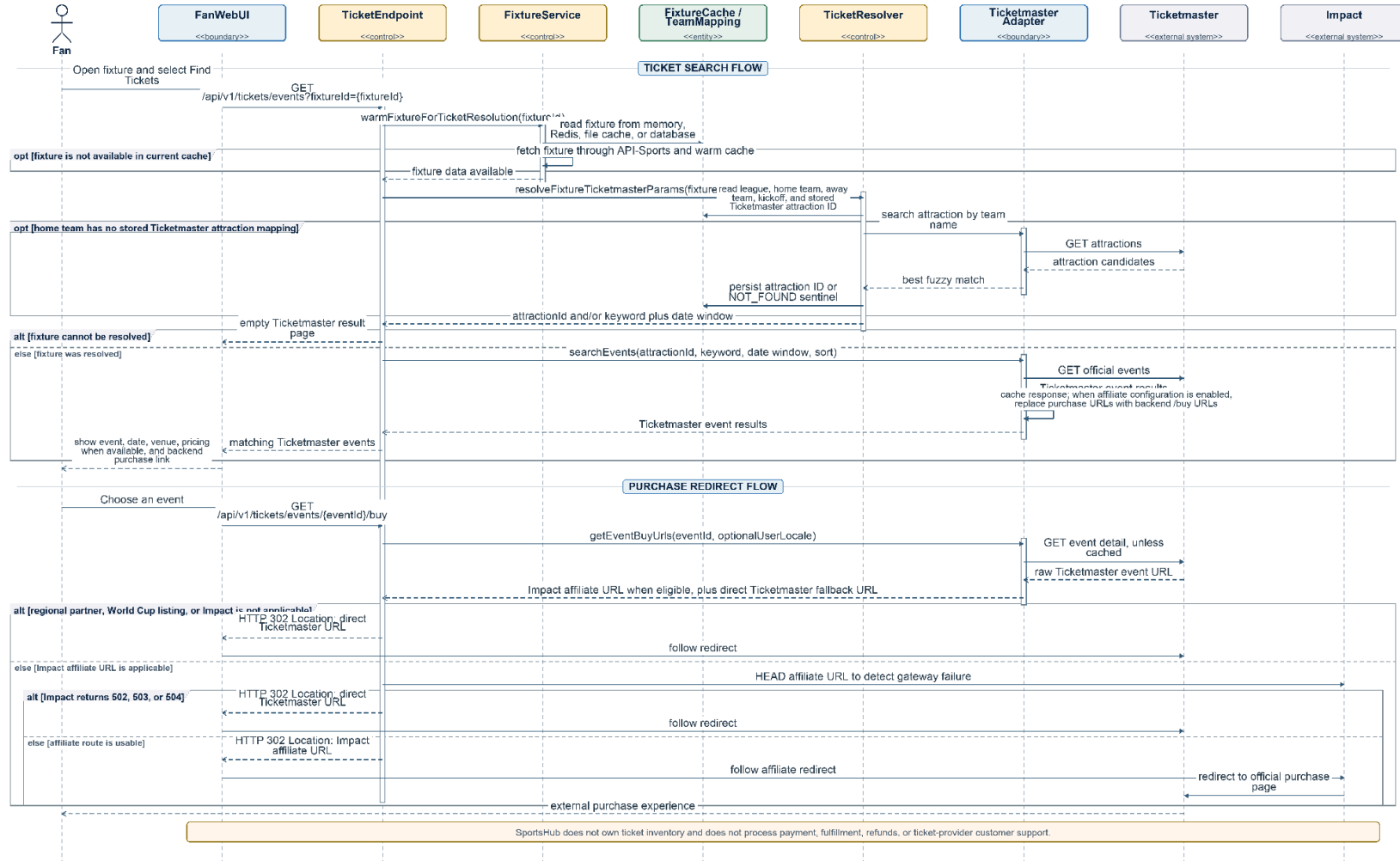


Figure 3. UML sequence diagram for discovering official tickets.

Analysis classes: Boundary – FanWebUI and TicketmasterAdapter; Control – TicketEndpoint, FixtureService, and TicketResolver; Entity – FixtureCache and TeamMapping; External systems – Ticketmaster and Impact.